P H Y S I C S

PHYS 00s

PHYS 10 SEM 0.00

Course ID: 009328

Physics Seminar

This seminar brings together physics students in all years to hear invited speakers, view physics-related films, and learn about current research. [Offered: F,W]

PHYS 100s

PHYS 111 LEC,TST,TUT 0.50

Course ID: 007388

Physics 1

An introduction to physics for students intending to concentrate their further studies in biology, dentistry, medicine and paramedicine; includes particle kinematics and dynamics, energy and momentum conservation, and rotational mechanics. [Offered: F, W; also offered online: W]

Antireq: PHYS 115, 121

Also offered Online

PHYS 111L LAB 0.25

Course ID: 007389

Physics 1 Laboratory

For students who have taken or are taking PHYS 111.

[Note: Lab alternates weeks. Offered: F]

Coreq: PHYS 111

Antireq: PHYS 121L or 131L

PHYS 112 LEC,TST,TUT 0.50

Course ID: 007390

Physics 2

A continuation of PHYS 111; includes simple harmonic motion, electrostatic force and potential, electric current and power, DC circuits, magnetic field and induction, wave motion, sound and optics. [Offered: W,S; also offered online: S]

Prereq: PHYS 111 or 121;

Antireq: PHYS 122, 125

Also offered Online

PHYS 112L LAB 0.25

Course ID: 007391

Physics 2 Laboratory

For students who have taken or are taking PHYS 112.

[Note: Lab alternates weeks. Offered: W,S]

Coreq: PHYS 112

Antireq: PHYS 122L or 132L

PHYS 115 LEC,TST,TUT 0.50

Course ID: 007392

Mechanics

Brief review of kinematics. Particle dynamics, work, energy, conservation of energy. Conservation of linear momentum, collisions, rotational kinematics and dynamics, conservation of angular momentum. Equilibrium of rigid bodies. [Offered: F, W, S]

Antireq: PHYS 111, 121; First year Engineering students only

PHYS 121 LEC,TST,TUT 0.50

Course ID: 007393

Mechanics

An introductory course in physics for students intending to concentrate their future studies in the physical sciences, optometry, or mathematics; includes vectors (dot and cross products), particle kinematics and dynamics, forces in nature, work and energy, conservation of energy and linear momentum, rotational kinematics and dynamics, and conservation of angular momentum.

[Note: Successful completion of 4U Calculus and Vectors, 4U Advanced Functions and 4U Physics is required. Offered: F, also offered online: W]

Coreq: One of MATH 104, 127, 137, 147.

Antireq: PHYS 111, 115, ECE 105

Also offered Online

PHYS 121L LAB 0.25

Course ID: 007394

Mechanics Laboratory

For students who have taken or are taking PHYS 121. [Offered: F]

Coreq: PHYS 121
Antireq: PHYS 111L or 131L

PHYS 122 LEC,TST,TUT 0.50

Course ID: 013651

Waves, Electricity and Magnetism

Simple harmonic motion, resonance, damped harmonic motion, complex numbers, wave motion and sound, electrostatic force and potential, electric current and power, capacitors, DC circuits, LRC circuits, introduction to magnetic fields Lorentz Force. [Offered: W, S; also offered online: S]

Prereq: One of PHYS 111 (minimum grade 70%), 115, 121, ECE 105.
Coreq: One of MATH 127, 137, 147.
Antireq: PHYS 112, 125
Also offered Online

PHYS 122L LAB 0.25

Course ID: 007396

Waves, Electricity and Magnetism Laboratory

For students who have taken or are taking PHYS 122. [Offered: W]

Coreq: PHYS 122.
Antireq: PHYS 112L or 132L

PHYS 124 LEC,TST,TUT 0.50

Course ID: 013640

Modern Physics

An introductory course in modern physics; includes relativity, quantum physics, atomic physics, nuclear physics, particle physics and gravitation. [Offered: W]

Prereq: One of PHYS 111 (minimum grade 70%), 115, 121, ECE 105.
Coreq: PHYS 112 or 122

PHYS 125 LEC,TST,TUT 0.50

Course ID: 007398

Physics for Engineers

Oscillations; simple harmonic motion. Wave motion, travelling and standing waves; transverse and longitudinal waves, including sound. Geometrical optics; reflection and refraction. Physical optics; interference and diffraction. Quantum physics; quantization of radiation; hydrogen atom. [Offered: W,S]

Prereq: PHYS 115; Engineering students only.

Antireq: PHYS 112, PHYS 122

PHYS 160L LAB 0.25

Course ID: 016093

Introductory Measurement Laboratory

A lab for students following the Honours Physics Plan.

[Note: Lab alternates weeks. Offered: W]

Prereq: Honours Physics students only.

Coreq: PHYS 122L.

Antireq: PHYS 132L

PHYS 175 LEC 0.50

Course ID: 013959

Introduction to the Universe

A survey course in astrophysics intended for Physics and Astronomy students. Astrophysical processes, the sky, the Sun, stars, black holes, the Milky Way and other galaxies, Big Bang cosmology.

[Note: students in Honours Physics and Astronomy programs must also take PHYS 175L; Offered: W]

Prereq: One of PHYS 111,115,121; one of MATH 116, 117, 127, 137,147.

Antireq: SCI 237, 238

PHYS 175L LAB 0.25

Course ID: 013960

Introduction to the Universe Laboratory

For students who are taking PHYS 175. [Offered: W]

Coreq: PHYS 175

PHYS 200s

PHYS 225 LEC,TUT 0.50

Course ID: 013968

Modeling Life Physics

Introduction to modeling living systems and their components. Statistical methods in data analysis, curve fitting, including p values. Fourier series and transforms, structural analysis, including nearest neighbor distributions in biomedical applications. Introduction to methods for analysis of transport properties in biological systems. Use of computers in these areas. [Offered: F]

Prereq: One of PHYS 112, 122; One of MATH 116, 117, 127, 137, 147.

Coreq: One of MATH 118, 119, 128, 138, 148

PHYS 232L LAB 0.25

Course ID: 010361

Measurement Laboratory

A laboratory that teaches programming for the computer interfacing of physics experiments and automatic data collection.

[Note: Lab alternate weeks. Offered: F].

Prereq: PHYS 122L; Honours Materials and Nanosciences and Honours Physics students only

PHYS 233 LEC,TUT 0.50

Course ID: 013969

Introduction to Quantum Mechanics

Introduction to quantization, wave-particle duality and the uncertainty principle The Schroedinger equation and solvable examples. Topics will include stationary states of particle-in-a-box, harmonic oscillator, and the hydrogen atom. Quantization of angular momentum and spin. Introduction to approximation methods including time-independent perturbation theory. Modern applications of quantum mechanics. [Offered: W]

Prereq: One of PHYS 112, 122, 125, ECE 106; One of MATH 118, 119, 128, 138, 148.

Coreq: One of MATH 211, 213, 228, AMATH 250, 251 ECE 205.

Antireq: CHEM 356, NE 232, PHYS 234, ECE 405

PHYS 234 LEC,TUT 0.50

Course ID: 007407

Quantum Physics 1

Background of quantum physics. Introduction to formalism of quantum physics. Introduction to operators. Quantization, waves, and particles. The uncertainty principle. The Schroedinger equation for one-dimensional problems: bound states in square wells, harmonic oscillator, transmission through barriers.

[Note: CS 114, PHYS 236, or knowledge of computational methods is recommended. Offered: W, S]

Prereq: PHYS 112 or 122; One of PHYS 249, MATH 114, 136; One of MATH 128, 138, 148.

Coreq: One of MATH 228, AMATH 250, AMATH 251.

Antireq: CHEM 356, NE 232, PHYS 233, ECE 405

PHYS 242 LEC,TUT 0.50

Course ID: 013643

Electricity and Magnetism 1

Electrostatics in vacuum, electric potential, conductors and currents, magnetostatics in vacuum, electromagnetic induction. [Offered: W,S]

Prereq: One of PHYS 112,122; One of MATH 128, 138, 148; (MATH 227 or co-requisite: AMATH 231).

Coreq: PHYS 242L for science students except Mathematical Physics & Materials and Nanosciences plan.

PHYS 242L LAB 0.25

Course ID: 007418

Electricity and Magnetism Laboratory

For students who have taken or are taking PHYS 242.

[Note: Lab alternates weeks. Offered: W,S]

Coreq: PHYS 242

Antireq: PHYS 224L

PHYS 249 LAB,LEC,TUT 0.50

Course ID: 016205

Computational Physics and Linear Algebra

Computational techniques; numerical accuracy and speed; Python libraries and implementation; algorithms for numerical integration; matrix operations. Computational approach to vectors in 2- and 3-space; linear equations, matrices, determinants; eigenvalues and diagonalization. [Offered: F]

Prereq: One of CS 114, 116, 136, 146

PHYS 256 LEC,TUT 0.50

Course ID: 007422

Geometrical and Physical Optics

Geometrical optics: image formation, ray tracing through multiple optical components, dispersion by prisms, optical fibers, optical instruments - eyes, telescopes, microscopes, and cameras, introduction to aberrations; physical optics: interference and interferometers, diffraction, imaging resolution, diffraction gratings and their use in spectroscopy; wave-particle duality; introduction to the electromagnetic nature of light and polarization. [Offered: F]

Prereq: PHYS 112 or 122; One of MATH 108, 128, 138, 148.

Coreq: PHYS 256L for Science students except for Mathematical Physics Plan.

Antireq: ECE 404

PHYS 256L LAB 0.25

Course ID: 007423

Optics Laboratory

For students who have taken or are taking PHYS 256.

[Note: Lab alternates weeks. Offered: F]

Coreq: PHYS 256

PHYS 260L LAB,LEC 0.25

Course ID: 013118

Intermediate Physics Laboratory

Experiments in selected physics topics.

[Note: Lab alternates weeks. Approximately a week before a particular lab is held, a one-hour lecture will be given on the material to be covered in the lab. Students in programs joint with Physics should see department for enrolment access. Offered: W,S]

Prereq: PHYS 232L or 256L; Honours Physics, Physics and Astronomy, or Materials and Nanosciences.

Coreq: PHYS 242

PHYS 263 LEC,TUT 0.50

Course ID: 003320

Classical Mechanics and Special Relativity

Newtonian dynamics of particles and systems of particles. Oscillations. Gravity and the central force problem. Lorentz transformations and relativistic dynamics. [Offered: W,S]

Prereq: One of PHYS 112, 122; One of MATH 108, 128, 138, 148; One of MATH 228, AMATH 250, 251;

Antireq: AMATH 261, 271

PHYS 267 LEC,TUT 0.50

Course ID: 015985

Probability, Statistics, and Data Analysis for Physics and Astronomy

Probability, probability distributions, errors, descriptive statistics, statistical inference (hypothesis testing, fitting, confidence intervals), computational methods (e.g., Monte Carlo) in Python, examples from physics and astronomy. [Offered: W, S]

Prereq: PHYS 122, MATH 227; One of PHYS 236 or CS 114, 116, 136, 146.

Antireq: STAT 230, 231

PHYS 270L LAB 0.25

Course ID: 013963

Astronomical Observations, Instrumentation, and Data Analysis Laboratory

Selected experiments in Astronomy. [Offered: W,S]

Prereq: PHYS 256 or ECE 404

PHYS 275 LEC 0.50

Course ID: 007428

Planets

Terrestrial and gas giant planets in the Solar System, asteroids and comets. Extrasolar planets and astrobiology. Star and planet formation. [Offered: F]

Pereq: One of PHYS 111, 115, 121, ECE 105

PHYS 280 LEC,TUT 0.50

Course ID: 012773

Introduction to Biophysics

Introduction to a physical understanding of biological systems at macro and molecular scales. The course is intended for second-year science and engineering students and will cover a broad spectrum of topics in biophysics, as well as an introduction to neurobiology, nanotechnology, and biotechnology. [Offered: W,S]

Prereq: Level at least 2A Honours Science or Engineering plans

(Cross-listed with BIOL 280)

PHYS 300s

PHYS 334 LEC,TUT 0.50

Course ID: 007434

Quantum Physics 2

Formalism of quantum mechanics. Operator approach to the harmonic oscillator. Quantum mechanics in three dimensions: Hydrogen atom, angular momentum and spin. Time-independent perturbation theory. Fine structure of hydrogen. Zeeman effect. Identical particles. The variational principle. Ground state of the helium atom. Applications in atomic and molecular physics. [Offered: W]

Prereq: PHYS 234 or CHEM 356; One of MATH 228, AMATH 250, 251; MATH 227 or 237 or 247.

Antireq: AMATH 373

PHYS 335 LEC 0.50

Course ID: 012032

Condensed Matter Physics

Overview of condensed matter ordered and disordered systems. Thermodynamic origin of order and phase transitions. Waves. Properties of the solid state. Crystals and fractals. Overview of Fourier Series. Reciprocal lattice. Diffraction. Classical elastic theory of the crystalline state. Electrons in a periodic potential, Band structure and Fermi surface. [Offered: W]

Prereq: PHYS 234 or CHEM 356; Level at least 3A Engineering, Mathematics or Science students

PHYS 342 LEC,TUT 0.50

Course ID: 013644

Electricity and Magnetism 2

Electric and magnetic fields in media, auxiliary fields, Maxwell's equations, electromagnetic waves, electric and magnetic properties of matter.

[Note: PHYS 236 or knowledge of computational methods recommended. Offered: F,S]

Prereq: PHYS 242; One of MATH 228, AMATH 250, AMATH 251.

PHYS 349 LAB,LEC,TUT 0.50

Course ID: 016206

Advanced Computational Physics

Algorithms for solving differential equations; Monte Carlo techniques; Fourier transforms; programming and computational techniques using Python, applied to physical problems such as astrophysics, electricity and magnetism, classical and quantum mechanics; introduction to machine learning and artificial intelligence. [Offered: W]

Prereq: PHYS 236 or PHYS 249; PHYS 234, PHYS 242, and (AMATH 271 or PHYS 263).

Antireq: PHYS 239

PHYS 358 LEC,TUT 0.50

Course ID: 007444

Thermal Physics

Temperature and thermodynamic equilibrium. Work, internal energy and heat; first law, with examples. Kinetic theory of gases. Basic probability theory. Microscopic states and entropy. Absolute temperature, reversibility and the second law. Thermodynamic Functions and Maxwell's relations. Phase transitions. Third Law. Other applications of thermodynamics. [Formerly PHYS 258. Offered: F, S]

Prereq: PHYS 112 or 122; One of MATH 227, 237, 247; One of MATH 228, AMATH 250, 251.

Antireq: CHEM 254, ECE 403

PHYS 359 LEC,TUT 0.50

Course ID: 007445

Statistical Mechanics

Fundamental postulate of statistical thermodynamics. Entropy. Microcanonical, canonical, and grand canonical ensembles. Fermi-Dirac, Bose-Einstein, and Boltzmann Statistics. Maxwell-Boltzmann velocity distribution. Applications to specific heat of solids, classical and quantum gases, electrons in metals, Planck's law of radiation, and Bose-Einstein condensation. [Offered: W]

Prereq: One of PHYS 358, ECE 403, CHEM 254, ME 250; One of PHYS 233, 234, CHEM 356, co-requisite: AMATH 373

PHYS 360A LAB 0.25

Course ID: 007446

Modern Physics Laboratory 1

Selected experiments in mechanics, optics, electronics, acoustics, atomic, molecular, nuclear, and solid state physics.

[Note: Students in programs joint with Physics should see department for enrolment access. Offered: F,W,S]

Prereq: One of MNS 201L, PHYS 260L, PHYS 270L; Honours Physics, Physics and Astronomy, Materials and Nanosciences or Mathematical Physics.

PHYS 360B LAB 0.25

Course ID: 007447

Modern Physics Laboratory 2

Continuation of PHYS 360A. [Offered: F,W,S]

Prereq: PHYS 360A

PHYS 363 LEC,TUT 0.50

Course ID: 007449

Intermediate Classical Mechanics

Non-inertial frames of reference. Calculus of variations. Lagrangian mechanics. Coupled oscillations and normal modes. Hamiltonian dynamics. [Offered: F,S]

Prereq: One of PHYS 236, CS 114, 116, 136, 146; One of PHYS 263 or AMATH 271; One of MATH 227, 237, 247; One of MATH 228, AMATH 250, AMATH 251

PHYS 364 LEC,TUT 0.50

Course ID: 007450

Mathematical Physics 1

Sturm-Liouville theory. Legendre, Bessel, and other special functions. Fourier series and introduction to Fourier transforms. Separation of variables. [Offered: F, S]

Prereq: MATH 227; One of PHYS 236, CS 114, 116, 136, 146; One of MATH 228, AMATH 250, 251; Honours Physics, Chemical Physics, Physics and Astronomy, Life Physics or Materials and Nanosciences students only.

Antireq: AMATH 353

PHYS 365 LEC,TUT 0.50

Course ID: 007451

Mathematical Physics 2

Complex variables, Cauchy-Riemann conditions, Cauchy integral formula, Taylor and Laurent expansions, residue theorem, contour integrals and applications. Fourier and Laplace transforms with applications. Green's functions. [Offered: W]

Prereq: MATH 227; One of MATH 228, AMATH 250, 251; Honours Physics, Chemical Physics, Physics and Astronomy, Life Physics and Materials and Nanosciences students only.

Antireq: AMATH 332

PHYS 370L LAB 0.25

Course ID: 013978

Astronomy Laboratory 1

Selected experiments in astronomy. [Offered: W]

Prereq: PHYS 267, 270L

PHYS 375 LEC,TUT 0.50

Course ID: 007457

Stars

Stellar distances, masses, ages. Stellar interiors and atmospheres, star formation and evolution. Supernovae, white dwarfs, neutron stars, black holes. [Offered: W]

Prereq: PHYS 112 or 122; One of PHYS 236, CS 114, 116, 136, 146; Two of PHYS 234, 242, 256, 275, 358, AMATH 271

PHYS 380 LEC 0.50

Course ID: 007458

Molecular and Cellular Biophysics

Cell structure and molecular composition; intermolecular interactions and hydration; protein structure and function; cytoskeletal filaments; DNA structure, packing and chromosomes; rate equations and biological dynamics (e.g., cytoskeletal polymerization); self-assembly; cell membranes; action potentials and biological electricity; molecular motors; cell motility.

[Note: Recommended PHYS 280/BIOL 280. Offered: F,S]

Prereq: PHYS 112 or 122; CHEM 123 or 125

PHYS 383 LEC,TUT 0.50

Course ID: 013970

Medical Physics

Applications of physics in medicine. The course will address basic concepts of medical imaging, nuclear medicine and radiation isotopes, radiation therapy and biomedical laser applications. Nuclear structure and binding energy. Nuclear decays, radioactivity and nuclear reactions. Interaction of radiation with matter. [Offered: W]

Prereq: PHYS 112 or 122; one of MATH 118, 119, 128, 138, 148; PHYS 280/BIOL 280

PHYS 391 LEC 0.50

Course ID: 007438

Electronics

Norton and Thévenin equivalent circuits, bipolar junction and field-effect transistors, operational amplifiers, negative feedback, noise, common circuits used for measurement and control of laboratory experiments, introduction to digital circuits. [Offered: W]

Prereq: One of PHYS 122 (winter 2019 or later), 224, 242.

Coreq: PHYS 391L

Also offered Online

PHYS 391L LAB 0.25

Course ID: 007439

Electronics Laboratory

For students who have taken or are taking PHYS 391.

[Note: Lab alternates weeks. Offered: W]

Coreq: PHYS 391

PHYS 393 LEC 0.50

Course ID: 013645

Physical Optics

Fourier Optics: diffraction, dispersion, coherence, imaging. Polarization: birefringence, Jones vectors, Mueller Matrices. Fresnel Equations: optics at an interface, thin films. [Offered: W even years]

Prereq: PHYS 256 or ECE 404; PHYS 364 or AMATH 353

PHYS 394 LEC 0.50

Course ID: 013646

Light-Matter Interactions

Lasers: semi-classical interaction, properties, cavity, Gaussian beams. Radiation and Detectors. Introduction to nonlinear optics. [Offered: W odd years]

Prereq: PHYS 256 or ECE 404; PHYS 364 or AMATH 353

PHYS 395 LEC,TUT 0.50

Course ID: 013659

Biophysics of Therapeutic Methods

The effect of radiation of various kinds on cells and tissues; elements of radiobiology and photobiology; molecular mechanisms of radiation-induced DNA damage and cell death, repair of radiation damage, dose-response relationships; tumor radiobiology and therapies, radiotherapy, photodynamic therapy, combination therapies; radiosensitizers and photosensitizers for cancer therapies; transdisciplinary advances in physical methods (ultrafast laser and ultrasound techniques) for biomedical applications.

[Note: PHYS 380 is recommended. Offered: W even years]

Prereq: One of PHYS 112, 122, 125, ECE 106; One of BIOL 130, 239, CHEM 123, 125, PHARM 141, PHYS 280/BIOL 280, PHYS 380; Level at least 3A Science, Mathematics or Engineering students

PHYS 396 LEC,SEM 0.50

Course ID: 011381

Biophysics of Imaging

Introduction to imaging concepts in biophysics with emphasis on the interrelationship between the physics principles of an imaging modality and the associated image reconstruction; methods for imaging at macroscopic to microscopic scales; computed tomography, magnetic resonance imaging, ultrasound, PET, optical imaging, optical and fluorescence microscopy, scanning probe microscopy (AFM, STM), optical tweezers, electron microscopy.

[Note: PHYS 380 recommended. Offered: W odd years]

Prereq: One of PHYS 112, 122, 125, ECE 106; One of MATH 128, 138, 148,(SYDE 111,112);One of PHYS 224, 233, 234, 242, 256, 280, 380, ECE 209, 370, 375, NE 232, 241, SYDE 283, AMATH 231, 373, CS 473, CHEM 209, 356; Level at least 3A SCI, MATH or ENG stdnts

PHYS 400s

PHYS 434 LEC 0.50

Course ID: 007463

Quantum Physics 3

Symmetries and conservation laws. Review of time-independent perturbation theory (degenerate and non-degenerate, Rayleigh-Schrodinger, Brillouin-Wigner and canonical perturbation theory; effective Hamiltonian derivation). Time-dependent perturbation theory (1st and 2nd order, adiabatic perturbation, Aharonov-Bohm effect). Fermi's golden rule. Two-level systems. Emission and absorption of radiation (applications). Second quantization of electromagnetic field in free space; photons. Spontaneous emission and natural lifetime; Lamb shift. Elements of scattering theory. Introduction to the Dirac equation. [Offered: F]

Prereq: PHYS 334 or AMATH 373; PHYS 364 or (AMATH 351 and 353)

PHYS 435 LEC 0.50

Course ID: 007464

Current Topics in Condensed Matter Physics

Physics pertaining to collective and emergent phenomena in condensed matter systems. Examples of topics to be covered include magnetism, superconductivity, heavy Fermion systems, quantum hall effect, protein folding, membranes, DNA physics, polymer physics, Modern experimental and theoretical techniques. [Offered: W]

Prereq: PHYS 335, 359; PHYS 334 or AMATH 373

PHYS 437A PRJ 0.50

Course ID: 007465

Research Project

A research project in any area of physics approved by the course co-ordinator(s).

[Note: Cumulative average must be 70% or higher; Offered: F,W,S]

Department Consent Required

Prereq: Level at least 4A Honours Physics, Chemical Physics, Mathematical Physics, Physics and Astronomy, Materials and Nanosciences, Life Physics or Biological and Medical Physics students only

PHYS 437B PRJ 0.50

Course ID: 007466

Research Project (Continued)

A research project in any area of physics approved by the course co-ordinator(s). This is often a continuation of the project started in PHYS 437A.

[Note: Cumulative average must be 70% or higher; Offered: F,W,S]

Department Consent Required

Prereq: PHYS 437A

PHYS 442 LEC 0.50

Course ID: 013647

Electricity and Magnetism 3

Review of the origin of Maxwell's equations, conservation laws, potential formulation of Maxwell's equations and gauge choices, vacuum solutions of Maxwell's equations (free space, waveguides, dispersion), solution to Maxwell's equations for arbitrary sources (static and time-dependent problems), relativistic formulation of electrodynamics, macroscopic Maxwell's equations and plane waves in macroscopic media. [Offered: F]

Prereq: PHYS 342; PHYS 364 or (AMATH 351 and AMATH 353)

PHYS 444 LEC 0.50

Course ID: 007470

Introduction to Particle Physics

This course introduces students to the standard model of particle physics. Topics covered include symmetries, particle classification, experimental methods and tools, scattering, Feynman diagrams, gauge theories, quantum electrodynamics, quarks, quantum chromodynamics, weak interactions, and the Higgs mechanism. [Offered: W]

Prereq: PHYS 334 or AMATH 373; PHYS 363; (PHYS 364 and 365) or (AMATH 332, 351, 353)

PHYS 449 LAB,LEC,TUT 0.50

Course ID: 016211

Machine Learning in Physics

Machine learning applications in the physical sciences. [Offered: F]

Prereq: PHYS 236 or PHYS 349

PHYS 454 LEC,TUT 0.50

Course ID: 003369

Quantum Theory 2

The Hilbert space of states, observables, and time evolution. Feynman path integral and Greens functions. Approximation methods. Co-ordinate transformations, angular momentum, and spin. The relation between symmetries and conservation laws. Density matrix, Ehrenfest theorem, and decoherence. Multiparticle quantum mechanics. Bell inequality and basics of quantum computing. [Offered: F]

Prereq: AMATH 373 or PHYS 334

(Cross-listed with AMATH 473)

PHYS 460A LAB 0.25

Course ID: 010364

Advanced Laboratory 1

Selected advanced experiments in mechanics, optics, acoustics, electronics, atomic, molecular, nuclear, and condensed matter physics. [Offered: F,W,S]

Prereq: PHYS 360A

PHYS 460B LAB 0.25

Course ID: 010365

Advanced Laboratory 2

A continuation of PHYS 460A. [Offered: F,W,S]

Prereq: PHYS 460A

PHYS 461 LEC 0.50

Course ID: 013648

Nanophysics

Fundamentals of nanotechnology. Applications of nanotechnology in biology and medicine. Nanotechnology and society. Physical foundations of nanodevices. Conduction at the nanoscale. Modern nanodevices. [Offered: F]

Prereq: PHYS 335 or ECE 231; Level at least 4A.

Antireq: NE 471

PHYS 467 LEC,TST 0.50

Course ID: 011497

Introduction to Quantum Information Processing

Basics of computational complexity; basics of quantum information; quantum phenomena; quantum circuits and universality; relationship between quantum and classical complexity classes; simple quantum algorithms; quantum Fourier transform; Shor factoring algorithm; Grover search algorithm; physical realization of quantum computation; error-correction and fault-tolerance; quantum key distribution.

Prereq: One of MATH 114, 115, 235, 245, PHYS 334; Level at least 3A

(Cross-listed with CS 467, CO 481)

PHYS 468 LEC 0.50

Course ID: 013649

Introduction to the Implementation of Quantum Information Processing

Photonic quantum computing, Superconducting qubits, NMR, Ion Trap quantum computing, Atomic quantum computing. [Offered: F]

Prereq: CO 481/CS 467/PHYS 467 or PHYS 334

PHYS 474 LEC 0.50

Course ID: 013964

Galaxies

Galaxy structure, formation and evolution. Dynamics and stellar populations; gas and dust; supermassive black holes; large-scale structure. [Offered: F]

Prereq: One of AMATH 261, 271, PHYS 263; PHYS 375; Level at least 4A Mathematics or Science

PHYS 475 LEC 0.50

Course ID: 007479

Cosmology

Robertson-Walker metric and Friedmann equations. Observational cosmology. Dark matter and dark energy. Gravitational lensing. Big Bang nucleosynthesis, the cosmic microwave background. Inflation. Structure formation.

[Note: PHYS 474 is recommended. Offered: F]

Prereq: One of AMATH 261, 271, PHYS 263; Level at least 4A in Mathematics or Science

PHYS 476 LEC,TUT 0.50

Course ID: 003371

Introduction to General Relativity

Tensor analysis. Curved space-time and the Einstein field equations. The Schwarzschild solution and applications. The Friedmann-Robertson-Walker cosmological models. [Offered: W]

Prereq: One of AMATH 261, 271, PHYS 263; AMATH 231 or MATH 227; Level at least 4A in Mathematics or Science

(Cross-listed with AMATH 475)

PHYS 483 LEC 0.50

Course ID: 013971

Advanced Therapeutic Concepts in Oncology and Medical Physics

This course examines advanced oncology therapeutic concepts required as a foundation for patient management in the inpatient, ambulatory, and community settings. Topics include a comprehensive review of cancer epidemiology and pathophysiology, therapeutic agents used in the oncology setting, management of cancer therapy toxicities, and treatment and palliation of cancer symptoms. Physics methods in radiation therapy and the medical physicist's role in cancer treatment will be addressed. [Offered: W; Held with PHARM 464]

Prereq: Level at least 4A Life Physics, Medical Physics Specialization

PHYS 484 LEC,TUT 0.50

Course ID: 015892

Quantum Theory 3: Quantum Information and Foundations

Theory of correlations and entanglement; theory of quantum channels, detectors; the measurement problem, in quantum mechanics; phase space formulation of quantum mechanics; entanglement in infinite dimensional quantum systems; introduction to open quantum systems; and exploration of current research directions in quantum information. [Offered: W]

Prereq: AMATH 473/PHYS 454; Level at least 4A Mathematics or Science students only

(Cross-listed with AMATH 474)

PHYS 490 LEC 0.50

Course ID: 010227

Special Topics in Physics

A lecture course offered in a particular branch of physics, subject to availability of instructor.

Prereq: Mathematical Physics, Physics, and Physics and Astronomy students only

PHYS 491 LEC 0.50

Course ID: 013972

Special Topics in Life, Medical and Biophysics

A lecture and project course offered in areas of life, medical and biophysics, which will add to the learning experience of the students in these plans at a fourth-year level. Lecture topics may include advanced topics in molecular and cellular biophysics, imaging, biophotonics, interaction of radiation with biomaterials, radiation dosimetry and other physics based diagnostics and therapeutics. Projects may include studies of recent advances in any areas of medical and biophysics. [Offered: F]

Prereq: Level at least 4A; one of PHYS 380, 383, 395, 396